

A Learning Rate Study

We find that training stitching adapters is relatively stable with AdamW. Our chosen learning rate 0.002 and weight decay 0.05 seem to be optimal across the various adapters and architectures that we checked. See Figure 7.

However, for fine-tuning all weights, we find larger performance differences across different learning rates (Figure 8). We also find the best learning rate is not consistent across all scenarios. Due to these issues, we run separate trials for all three learning rates and report the best performer in each unique scenario (see Section 2 subsection “Optimization”).

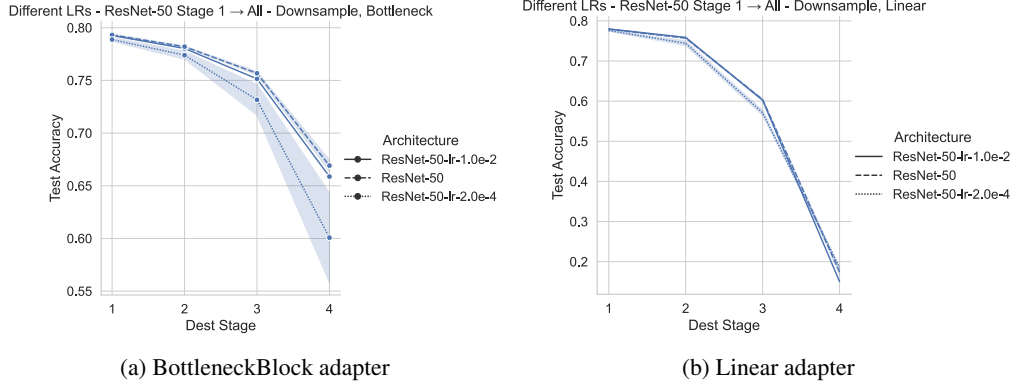


Figure 7: Optimizing adapters using various learning rates, to stitch ResNet-50 Stage 1 to others. We find that the default learning rate (the dashed line) is generally optimal, with minor differences to other learning rates.

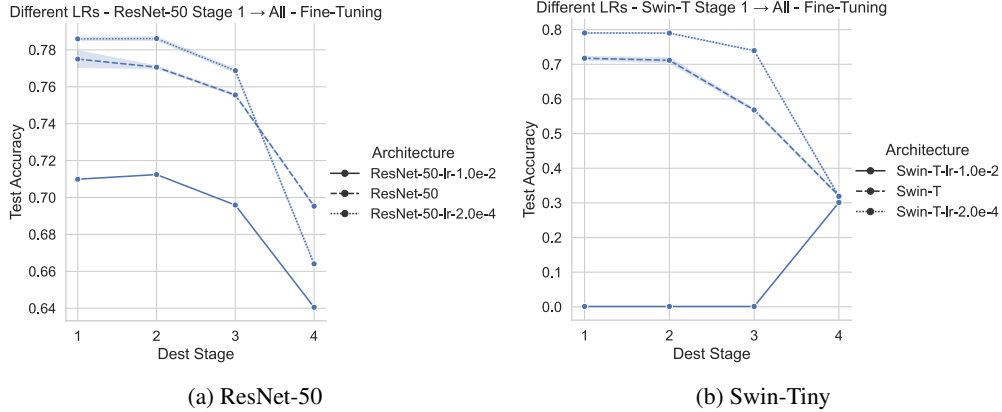


Figure 8: Fine-tuning all weights using various learning rates, on two different architectures. In contrast to the adapters, we find a larger performance difference across different learning rates, especially for Swin-T. We find that fine-tuning tends to prefer the smaller learning rate (the dotted line), but the best learning rate is not consistent across all scenarios. Due to these issues, we run separate trials for all three learning rates and report the best performer in each unique scenario.

B Full Results

See Figures 9, 10, 11 for accuracy vs. depth results across all scenarios. See Figures 12, 13, 14 for accuracy vs. FLOPs results across all scenarios.

Figure 15 shows a comparison of the best stitching results for each architecture, including the MobileNetV3 variants which were omitted from Figure 6 for visibility.

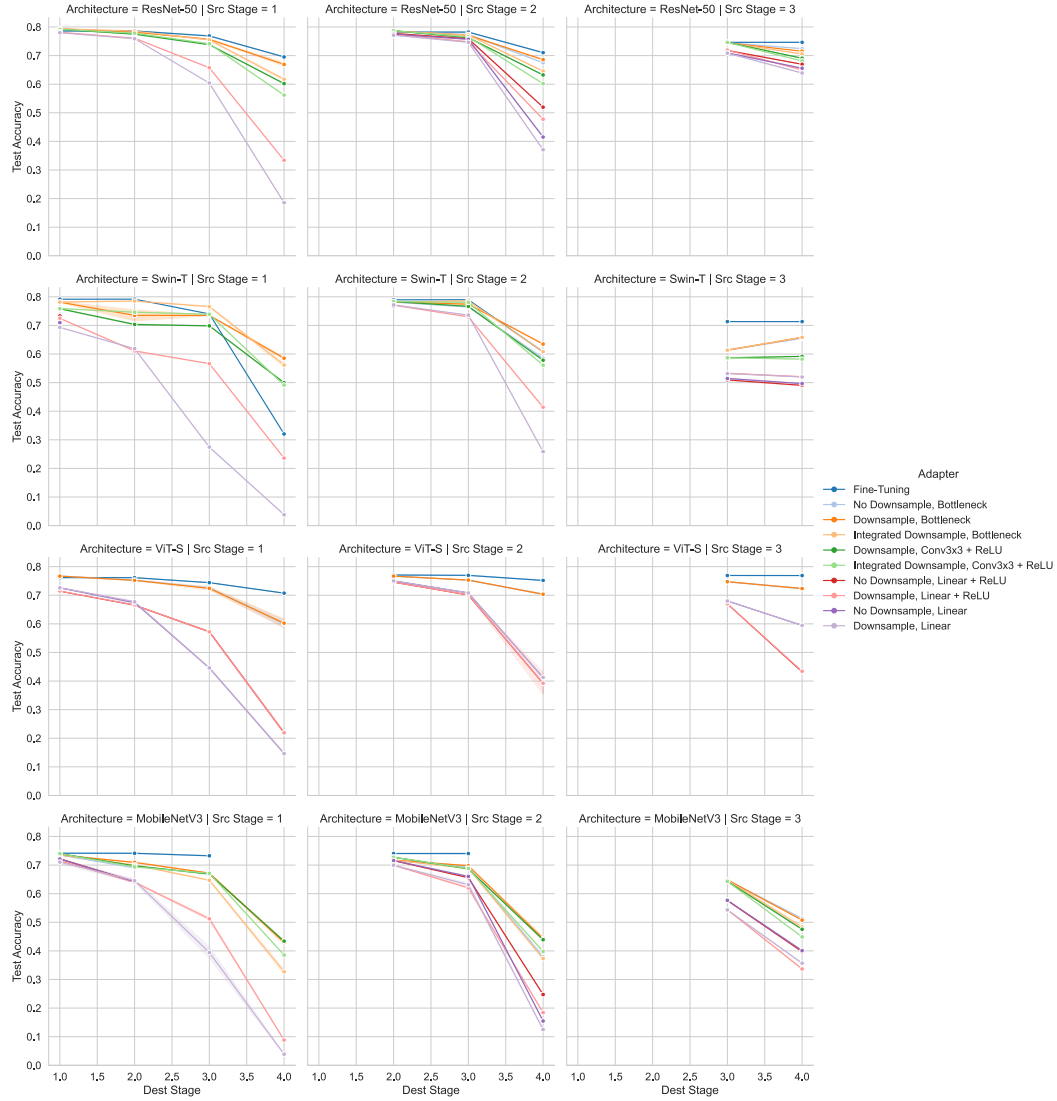


Figure 9: Accuracy vs. stitching gap size, across three stages and four architecture combinations. (Part 1) (Note: In the “Architecture = ViT-S” plots, the line named “Bottleneck” is actually the Vision Transformer Block, renamed to simplify the legend. This is the only place where ViT Blocks were used as adapters.)

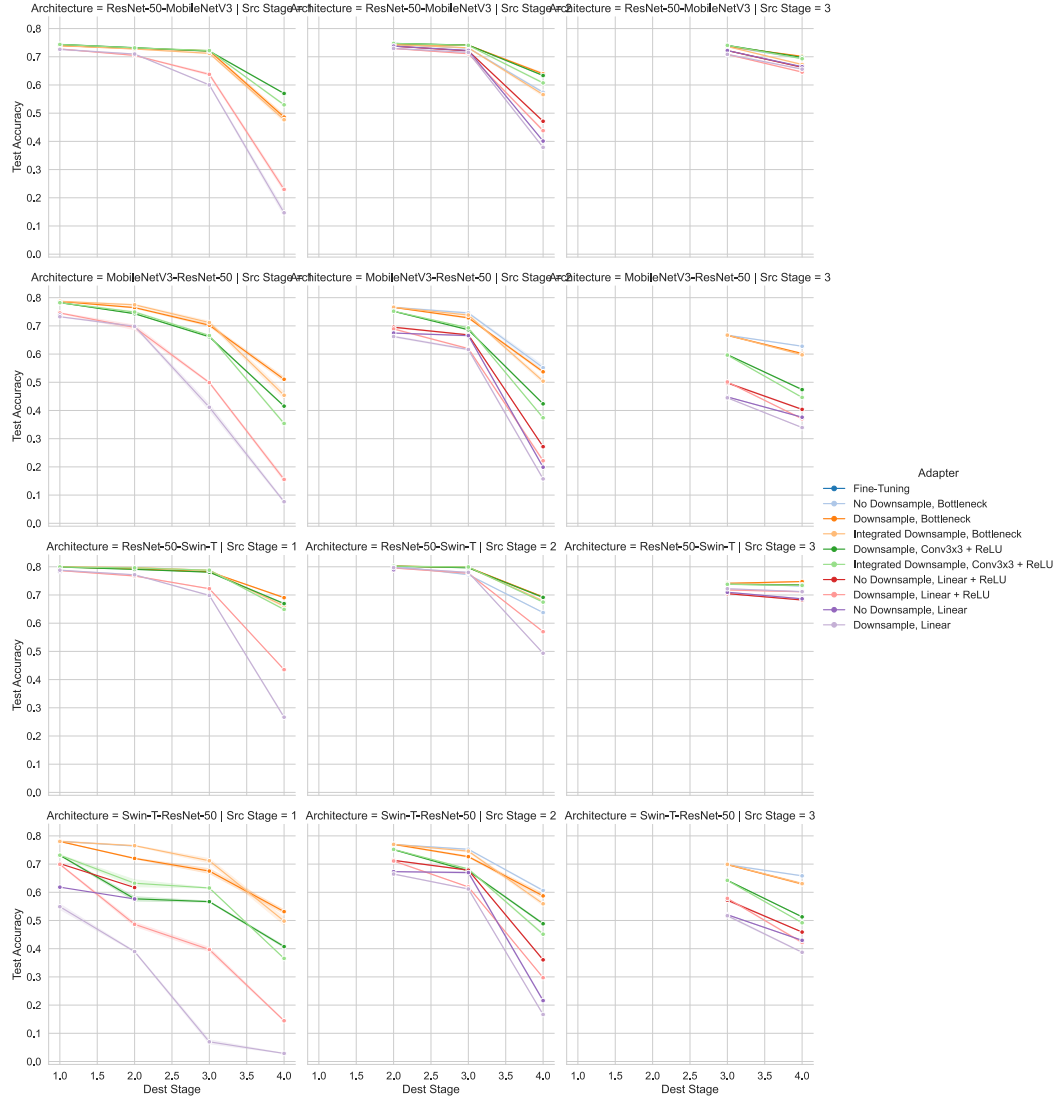


Figure 10: Accuracy vs. stitching gap size, across three stages and four architecture combinations. (Part 2)

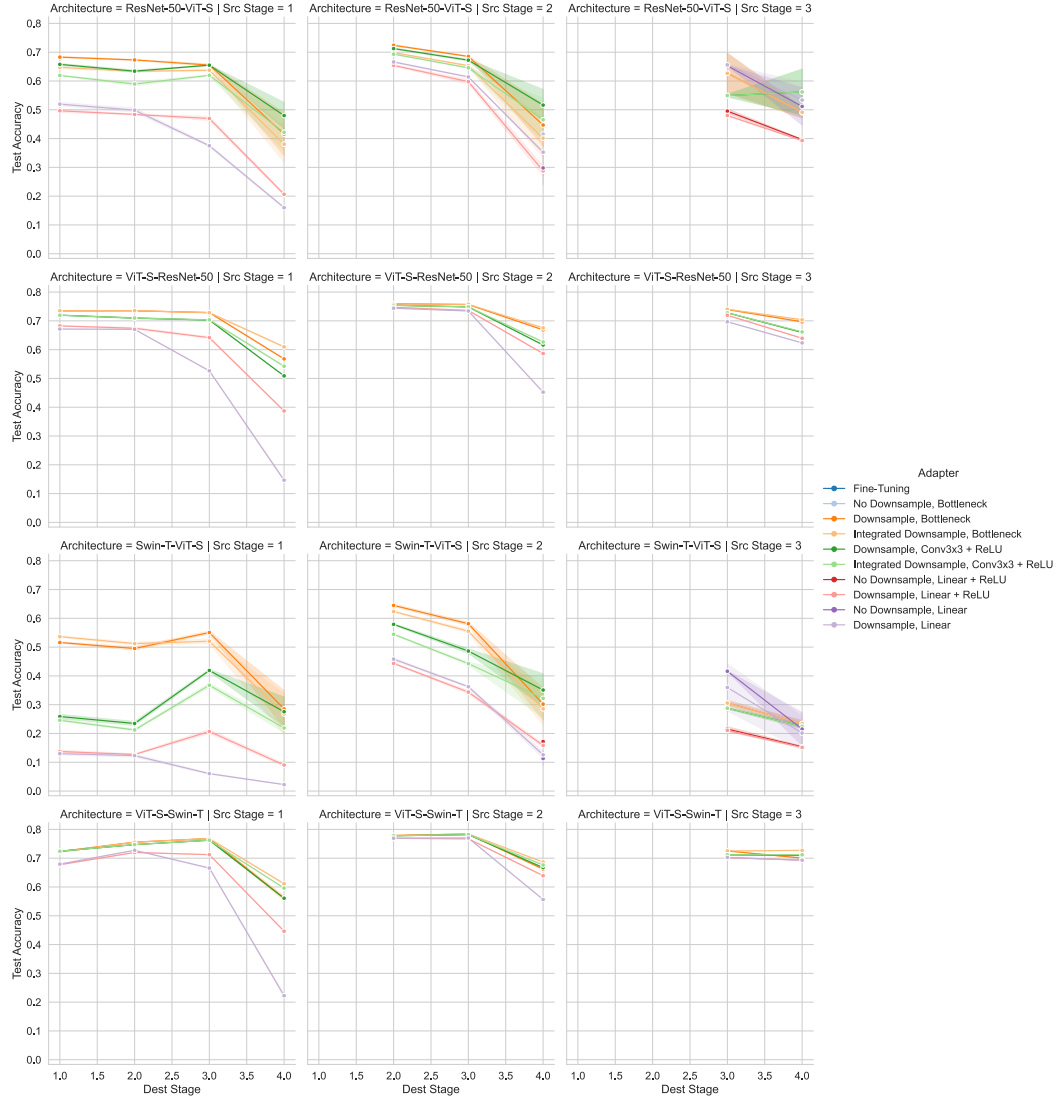


Figure 11: Accuracy vs. stitching gap size, across three stages and four architecture combinations. (Part 3)

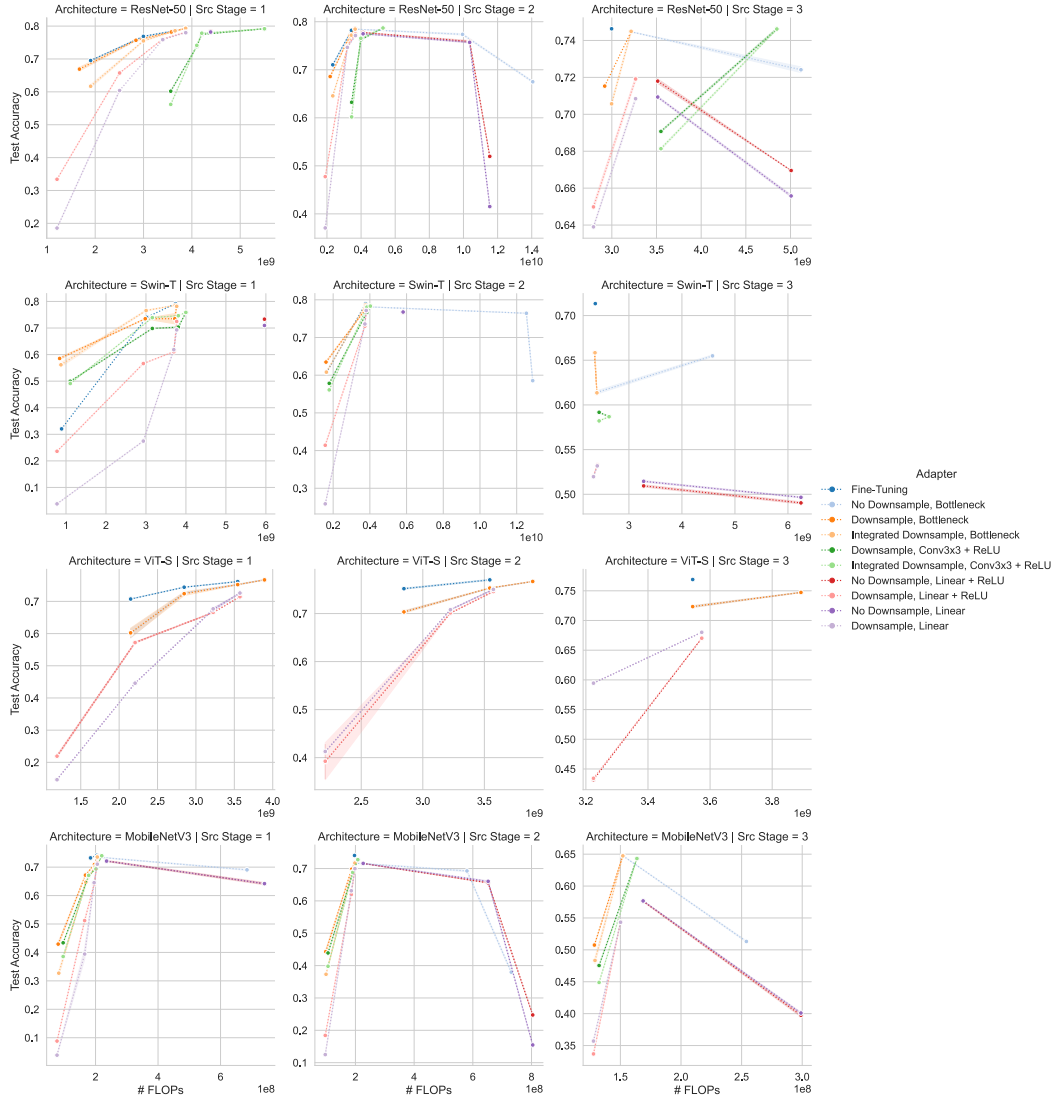


Figure 12: Accuracy vs. FLOP count of the resulting model, across three stages on all four architectures. (Part 1) (Note: In the “Architecture = ViT-S” plots, the line named “Bottleneck” is actually the Vision Transformer Block, renamed to simplify the legend. This is the only place where ViT Blocks were used as adapters.)

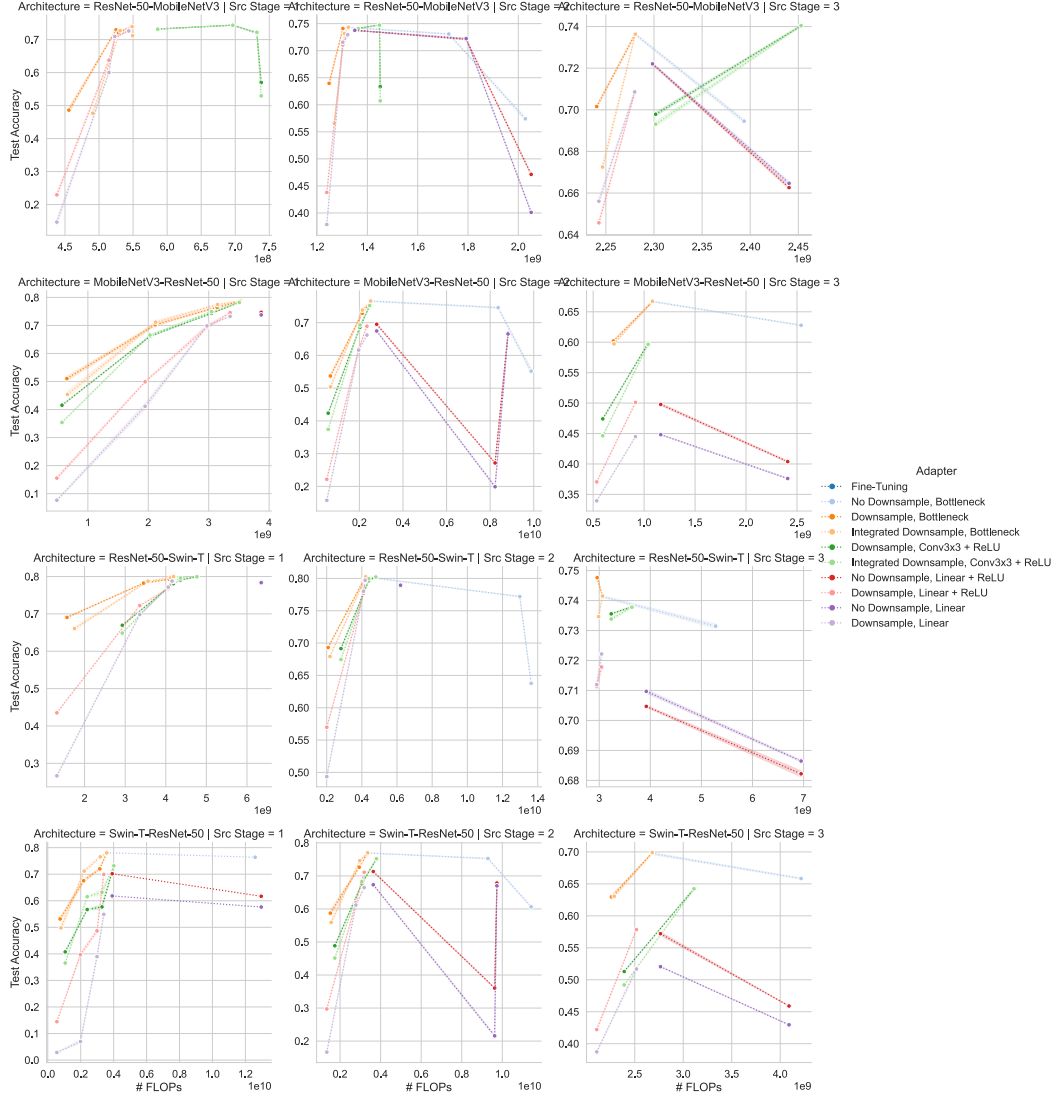


Figure 13: Accuracy vs. FLOP count of the resulting model, across three stages on all four architectures. (Part 2)

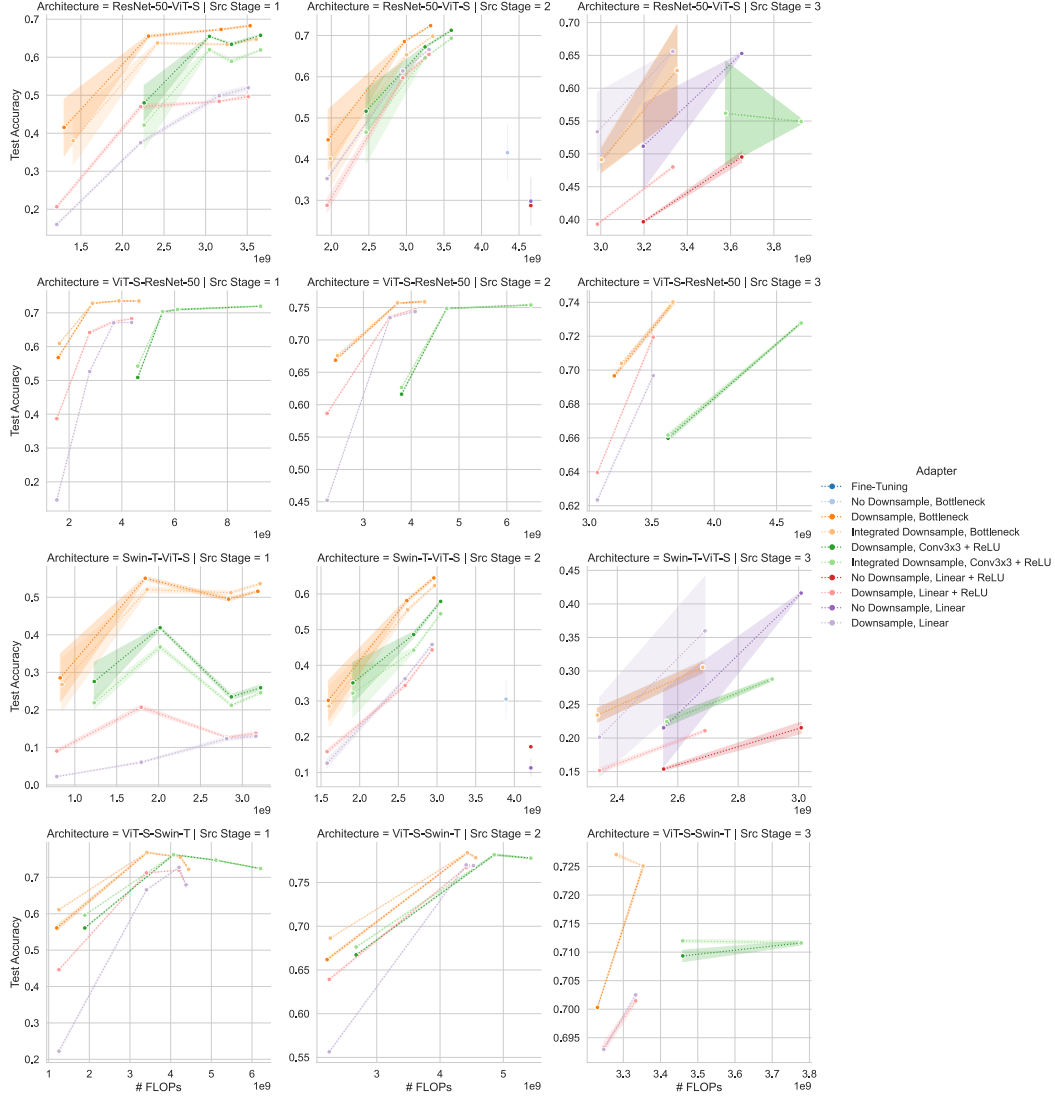


Figure 14: Accuracy vs. FLOP count of the resulting model, across three stages on all four architectures. (Part 3)

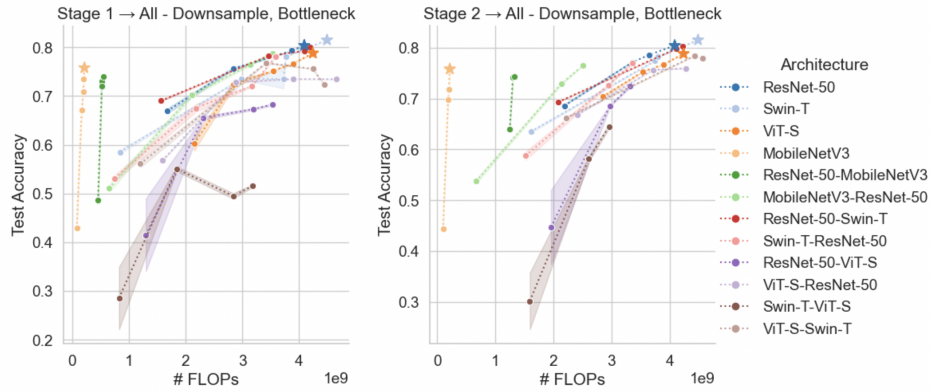


Figure 15: Accuracy vs. FLOP count, when stitching using full blocks as adapters. (Up and to the left means better accuracy with less compute. Stars denote the original accuracy/FLOPs of each source architecture. **Left:** Stitching Stage 1 to other stages. **Right:** Stitching Stage 2 to other stages.)